

Incident Report — MFA Fatigue: The Notification Flood

SOC Analysis & Triage · Easy Operation

1. Header / Metadata

Operation ID	OPS-2026-0412-05
Operation name	MFA Fatigue: The Notification Flood
Started (UTC)	2026-04-12 14:22:05
Closed (UTC)	2026-04-12 23:45:33
Analyst	Ndumiso Mbanjwa
Cohort	May26 Cybersecurity
Difficulty	Easy
Severity	Critical
Final verdict	True Positive
Status	Closed – Escalated for host isolation, credential/hash rotation and data-breach review

2. Executive Summary

On 2026-04-12, an employee named Nora Beckett, working on computer corp-wks-042 (10.0.2.219), received an email pretending to be a Dropbox file-sharing notice about “Q2 Revenue Forecast.xlsx”. It came from drbx-share[.]com, a name chosen to look like Dropbox. When she clicked, her browser was passed on to flowerstorm-auth[.]net, a fake sign-in page that sits in the middle between the victim and the real Microsoft service and copies whatever is typed. That gave the attacker her password, but not the second step of her login — the approval prompt on her phone.

To get past that, the attacker used “MFA fatigue”: from the address 198.88.230[.]252 they triggered six approval prompts on Nora's phone in about 90 seconds. She denied the first five and approved the sixth, most likely just to make the notifications stop. That single tap gave the attacker a fully signed-in session. They used it to drop a program called adversary-ai-agent.exe into her Temp folder, which connected back to the attacker's control server at 103.112.22[.]236. The attacker then used certutil.exe — a normal Windows tool, which helps the activity blend in — to download a portable copy of Python, and ran a well-known hacking script called wmiexec.py (from the Impacket toolkit) through python.exe. Instead of a password, the script used a stolen password “hash” (a scrambled form of the password that Windows will still accept) to log in to a second computer, WS-MKT-02.

Attempts on two other computers were refused. On WS-MKT-02 the attacker gathered marketing data and, over about six hours, sent 4.2 GB of it out to 103.112.22[.]236.

This is a confirmed True Positive: I resolved all 6 of the platform's scored tasks, and the verdict rests on the authentication timeline, the endpoint process tree, and the network logs, which together show every stage from the phishing email to the data theft.

3. Operation Details

Triggering detection	A Priority 1 alert: user nora.beckett on corp-wks-042 (10.0.2.219) showing signs of a multi-stage compromise.
Source	198.88.230[.]252 (where the MFA prompts and the stolen sign-in came from); 103.112.22[.]236 (the attacker's control and data drop-off server, also behind flowerstorm-auth[.]net)
Destination	nora.beckett's Microsoft/Azure AD account and workstation corp-wks-042; then internal workstation WS-MKT-02 (10.0.1.112)
User / Host affected	nora.beckett; corp-wks-042 (10.0.2.219) and WS-MKT-02 (10.0.1.112) compromised; WS-HR-01 and WS-SALES-01 targeted but access was denied
Artefact	Phishing email "Shared Document: Q2 Revenue Forecast.xlsx"; adversary-ai-agent.exe (unsigned, in the user's Temp folder); python-embed.zip fetched by certutil.exe; python.exe running wmiexec.py with a stolen NTLM hash
Direction	Inbound: phishing email and MFA push flood. Internal (lateral): corp-wks-042 → WS-MKT-02 over SMB port 445. Outbound: control traffic and 4.2 GB exfiltration to 103.112.22[.]236
Tool surfaces used	Security log platform (SIEM) and endpoint detection (XDR) — event timeline and process tree

4. Investigation & Triage

Task	Objective	What I checked	Finding	Decision & reasoning	Fig. ref
Task 1 – The phishing hook	Understand how the attack started	The case briefing	nora.beckett on corp-wks-042 (10.0.2.219) received a fake Dropbox email at 14:22, clicked at 14:25 and was sent	This gave the starting point and the two tools to correlate: SIEM for sign-in and network events, XDR for	Fig. 1

Task	Objective	What I checked	Finding	Decision & reasoning	Fig. ref
			to flowerstorm-auth.net, an adversary-in-the-middle proxy	what ran on the computer	
Task 2 – The MFA fatigue pattern	Name the technique shown in the authentication events	The XDR authentication timeline for nora.beckett, 09:00–14:30	Five MFA push prompts DENIED between 14:26:02 and 14:27:08 (about 16 seconds apart), one APPROVED at 14:27:31, then a token-based logon at 14:28:05 from 198.88.230.252 — not her workstation address from the morning logon	Repeated prompts in quick succession from an outside address until one is approved is MFA request flooding: MITRE ATT&CK T1621 (Multi-Factor Authentication Request Generation)	Fig. 2, 3, 4, 5
Task 3 – Identifying the attack tool	Find which process actually ran wmiexec.py	The XDR process tree for corp-wks-042, reading every child's full command line	adversary-ai-agent.exe (unsigned, from Temp, talking to c2.flowerstorm-auth.net:8443) spawned cmd.exe for reconnaissance, certutil.exe to download python-embed.zip, and python.exe running “wmiexec.py CORP/nora.beckett @10.0.1.112 - hashes ...”	The answer is python.exe. A normal, legitimate program is still the attack tool when its command line says so — detection has to look at command lines and parent processes, not just file names	Fig. 6, 7
Task 4 – Lateral movement discovery	Recover the credential the attacker replayed	The SIEM network logs from corp-wks-042, with the SMB row	At 14:58:14 python.exe connected to 10.0.1.112:445 (“SMB session via wmiexec.py, pass-	Because the attacker used the hash, not the password, a password reset alone is not enough: the credential	Fig. 8, 9

Task	Objective	What I checked	Finding	Decision & reasoning	Fig. ref
		expanded to its parsed fields	the-hash"); the CommandLine field shows -hashes aad3b435b51404e eaad3b435b51404 ee:ccef208a86548b 2a371ab851255ff0 93. The last row shows 4.2 GB leaving to 103.112.22.236	material itself must be invalidated and other hosts hunted	
Task 5 – Mapping the blast radius	Work out which machine was actually compromised	The XDR lateral-movement timeline	DNS lookups for four hosts; SMB/NTLM accepted on 10.0.1.112 (WS-MKT-02) followed by remote commands and file staging there; WS-HR-01 and WS-SALES-01 returned access denied	Only WS-MKT-02 shows attacker activity after the connection. Looking a host up is not compromising it, so WS-HR-01 and WS-SALES-01 are audited rather than isolated	Fig. 10
Task 6 – Containment and lessons learned	Confirm the full attack chain and the containment plan	The platform's attack chain summary and containment list	Chain confirmed end to end: Dropbox lure → AiTM proxy → MFA fatigue (T1621) → agent + C2 → certutil + python wmiexec.py over SMB (T1021.002) → 4.2 GB exfiltration	Isolate both hosts first to stop the data flow and the attacker's control channel, then revoke sessions and rotate the stolen hash	Fig. 11

5. Threat Intelligence

Indicator (defanged)	Tool	Result / score	Date checked (UTC)	Notes
flowerstorm-auth[.]net	VirusTotal	No record — VirusTotal returned “Item not found”	2026-09-15	This only means the domain has never been submitted to VirusTotal, not that it is safe. Our own logs show it acting as the AiTM phishing proxy and (as c2.flowerstorm-auth[.]net) the control channel.
drbx-share[.]com	VirusTotal	No record — “Item not found”	2026-09-15	Same as above: unknown to VirusTotal. The finding rests on the email and redirect logs showing it delivered the lure and redirected Nora to the proxy.
198.88.230[.]252	VirusTotal	0 out of 89 vendors flagged it; AS 2914 (NTT America), US	2026-09-15	A clean score is not proof of safety. The authentication timeline directly shows the MFA flood and stolen token logon coming from this address.
103.112.22[.]236	VirusTotal	0 out of 89 vendors flagged it; AS 137678 (Protoact Digital Network), IN	2026-09-15	Unflagged, but our logs show adversary-ai-agent.exe beaconing to it and 4.2 GB of data leaving to it — far stronger evidence than an outside reputation score.

This is assessed with high confidence as an MFA-fatigue identity attack delivered through the FlowerStorm adversary-in-the-middle phishing kit, followed by hands-on lateral movement with Impacket and data theft. None of the four network indicators is flagged on VirusTotal: the two domains have never been submitted at all, and both IP addresses score 0 of 89. That is not a

contradiction. Purpose-built phishing and control infrastructure is often short-lived and never reaches public reputation feeds, which is exactly why the platform describes this scenario as a case for behavioural analysis over simple indicator matching. A clean or missing VirusTotal result is treated as inconclusive, not as proof of safety — every finding here is backed by the organisation's own authentication, endpoint and network logs.

6. Timeline (UTC)

Time (UTC)	Event
2026-04-12 09:00:14	Nora's normal morning logon to corp-wks-042 using Windows Hello
2026-04-12 14:22:05	Phishing email "Shared Document: Q2 Revenue Forecast.xlsx" arrives from dropbox-notify@drbx-share[.]com
2026-04-12 14:25:18	High — Nora clicks; her browser is redirected drbx-share[.]com → flowerstorm-auth[.]net (AiTM proxy)
2026-04-12 14:26:02–14:27:08	Five MFA push prompts are sent to Nora and denied
2026-04-12 14:27:31	Critical — the sixth MFA prompt is approved
2026-04-12 14:28:05	Critical — token-based logon for nora.beckett from 198.88.230[.]252; attacker session established
2026-04-12 14:30:18	adversary-ai-agent.exe starts from Nora's Temp folder and connects to c2.flowerstorm-auth[.]net (103.112.22[.]236:8443)
2026-04-12 14:52:18	Reconnaissance: cmd.exe runs whoami, ipconfig and a Domain Admins group lookup
2026-04-12 14:53:45–14:55:02	DNS lookups for LAPTOP-ENG-03, WS-MKT-02, WS-HR-01 and WS-SALES-01
2026-04-12 14:56:40	certutil.exe downloads python-embed.zip from c2.flowerstorm-auth[.]net
2026-04-12 14:58:02–14:58:14	High — python.exe runs wmiexec.py with a stolen NTLM hash; SMB session to WS-MKT-02 (10.0.1.112) accepted
2026-04-12 15:00:42	High — remote file enumeration on WS-MKT-02
2026-04-12 15:02:28–15:03:44	SMB attempts to WS-HR-01 and WS-SALES-01 fail (access denied)
2026-04-12 15:05:18	High — marketing data staged in C:\Windows\Temp\staging\ on WS-MKT-02
2026-04-12 23:45:33	High — exfiltration to 103.112.22[.]236 completes: 4.2 GB over about 6 hours

7. Indicators of Compromise

Type	Indicator (defanged)	Context
Domain	drbx-share[.]com	Dropbox-lookalike sender and first redirect of the phishing link
Domain	flowerstorm-auth[.]net	FlowerStorm AiTM phishing proxy; c2.flowerstorm-auth[.]net also used for control and tool download
IPv4	198.88.230[.]252	Source of the MFA push flood and the stolen token-based logon
IPv4	103.112.22[.]236	Control (C2) server for adversary-ai-agent.exe and destination of the 4.2 GB exfiltration
File name	adversary-ai-agent.exe	Unsigned attacker agent in C:\Users\nora.beckett\AppData\Local\Temp\
Process	python.exe (running wmiexec.py)	Portable Python in the user's Temp folder used to run Impacket wmiexec.py for lateral movement
NTLM hash	aad3b435b51404eeaad3b435b51404ee:ccef208a86548b2a371ab851255ff093	LM:NT pair replayed over SMB (pass-the-hash) to reach WS-MKT-02
Hostname	WS-MKT-02 (10.0.1.112)	Second compromised workstation; data staged here before exfiltration

STIX 2.1 Bundle

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    "name": "FlowerStorm AiTM proxy / C2 domain",
    "description": "AiTM phishing proxy; subdomain c2.flowerstorm-auth.net used
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  "indicator_types": [

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C:\\Users\\nora.beckett\\AppData\\Local\\Temp\\. ",
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}
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8. MITRE ATT&CK Mapping

Tactic	Technique	ID
Initial Access	Phishing: Spearphishing Link	T1566.002
Credential Access	Adversary-in-the-Middle	T1557
Credential Access	Multi-Factor Authentication Request Generation	T1621
Execution	Command and Scripting Interpreter: Python	T1059.006
Discovery	Permission Groups Discovery: Domain Groups	T1069.002
Command and Control	Ingress Tool Transfer	T1105
Lateral Movement	Use Alternate Authentication Material: Pass the Hash	T1550.002
Lateral Movement	Remote Services: SMB/Windows Admin Shares	T1021.002
Collection	Data Staged: Remote Data Staging	T1074.002
Exfiltration	Exfiltration Over C2 Channel	T1041

T1621 is the platform's own Confirmed finding and T1021.002 is named in the platform's attack chain summary. The other techniques are mapped only from behaviour the platform shows as confirmed in its process tree, event timeline and attack chain summary; no technique is listed without that evidence.

9. Impact & Containment

Impact: the attacker took over Nora Beckett's account despite MFA, planted a remote-control program on her computer, stole reusable credential material (an NTLM hash), used it to break into a second workstation (WS-MKT-02), and stole 4.2 GB of marketing department data. Because data left the organisation, this is also a potential data-breach notification matter. Containment steps, in priority order:

1. Isolate corp-wks-042 (10.0.2.219) and WS-MKT-02 (10.0.1.112) from the network using XDR host isolation, so the attacker's control channel and any further data transfer are cut off immediately.
2. Block 198.88.230[.]252, 103.112.22[.]236, flowerstorm-auth[.]net (including c2.) and drbx-share[.]com at the firewall, proxy and DNS filter.
3. Revoke all active sessions and tokens for nora.beckett, then force a password reset and MFA re-enrollment.
4. Invalidate the stolen NTLM hash by rotating the credential, and check where else that account has access so the hash cannot be reused.
5. Hunt every endpoint for adversary-ai-agent.exe, python.exe running from Temp folders, certutil.exe downloads and wmiexec.py artefacts; audit WS-HR-01 and WS-SALES-01 to

confirm no other access path succeeded.

6. Notify the data protection team about the 4.2 GB exfiltration from the marketing department for possible regulatory reporting.

10. Conclusion & Recommendations

Final verdict: True Positive. One tap on an MFA prompt turned a phishing email into a full network intrusion: an account takeover, a remote-control agent, lateral movement with a stolen hash, and the theft of 4.2 GB of data. The evidence is complete and consistent across the authentication timeline, the process tree and the network logs. The key lesson is that MFA is not a silver bullet: simple push approvals can be worn down, so the defence has to make approval harder to give by mistake and make stolen sessions and hashes less useful.

Recommendations:

1. Turn on MFA number matching (the user must type a code shown on the login screen) or move to phishing-resistant sign-in such as passkeys, so a push flood cannot succeed.
2. Create a SIEM rule that locks the account and alerts the SOC when more than 3 MFA denials are followed by an approval within 5 minutes, especially from a non-corporate IP address.
3. Use Conditional Access to block sign-ins from unmanaged devices and unfamiliar networks.
4. Alert on certutil.exe downloading files and on python.exe (or other interpreters) launched from user Temp folders; restrict NTLM and workstation-to-workstation SMB to limit pass-the-hash.
5. Train staff to deny unexpected MFA prompts and report them to the SOC straight away.

11. Evidence / Screenshots

Figure	Caption	Referenced in
Figure 1	The operation header showing the completed 100% badge, 6/6 tasks, the scenario's title, difficulty (Easy), SIEM/XDR tool surfaces and points	Section 3, Task 1
Figure 2	The XDR event timeline: the fake Dropbox email from dropbox-notify@drbx-share.com at 14:22:05 and the High-severity redirect drbx-share.com → flowerstorm-auth.net (AiTM proxy detected) at 14:25:18	Task 1, Task 2
Figure 3	The authentication timeline: repeated MFA push prompts to nora.beckett DENIED, then APPROVED at 14:27:31 (Critical)	Task 2
Figure 4	The authentication timeline, final rows: the approval at 14:27:31 followed by a successful token-based logon from 198.88.230.252 that dropped adversary-ai-agent.exe beaconing to 103.112.22.236	Task 2
Figure 5	The Task 2 "Confirmed finding" panel: t1621 (Multi-Factor Authentication Request Generation)	Task 2, Section 8
Figure 6	The XDR process tree for corp-wks-042: adversary-ai-agent.exe (Suspicious) talking to c2.flowerstorm-auth.net:8443, with child processes cmd.exe (recon), python.exe running wmiexec.py against 10.0.1.112, and certutil.exe downloading python-embed.zip	Task 3
Figure 7	The Task 3 "Confirmed finding" panel: python.exe as the process that directly executed wmiexec.py	Task 3
Figure 8	The SIEM log row at 14:58:14 expanded to its parsed fields: CommandLine "python.exe wmiexec.py CORP/nora.beckett@10.0.1.112 - hashes aad3b435....:ccef208a...", destination 10.0.1.112, internal zone	Task 4
Figure 9	The SIEM network log's last rows: failed SMB attempt to 10.0.1.98 and, at 23:45:33,	Task 4, Section 9

Figure	Caption	Referenced in
	adversary-ai-agent.exe sending 4.2 GB over 6 hours to 103.112.22.236 (High)	
Figure 10	The XDR lateral-movement timeline: SMB access to WS-HR-01 and WS-SALES-01 FAILED (access denied), while WS-MKT-02 shows file staging to C:\Windows\Temp\staging\	Task 5
Figure 11	The platform's attack chain summary and immediate containment actions, confirming T1621, T1021.002, WS-MKT-02, the NTLM hash and the 4.2 GB exfiltration	Task 6, Section 9
Figure 12	VirusTotal lookup for flowerstorm-auth.net: "Item not found" (never submitted)	Section 5
Figure 13	VirusTotal lookup for drbx-share.com: "Item not found" (never submitted)	Section 5
Figure 14	VirusTotal results for 198.88.230.252: 0 of 89 vendors flag it (AS 2914, NTT America, US)	Section 5
Figure 15	VirusTotal results for 103.112.22.236: 0 of 89 vendors flag it (AS 137678, Protoact Digital Network, IN)	Section 5

Figure 1. The operation header showing the completed 100% badge, 6/6 tasks, the scenario's title, difficulty (Easy), SIEM/XDR tool surfaces and points.

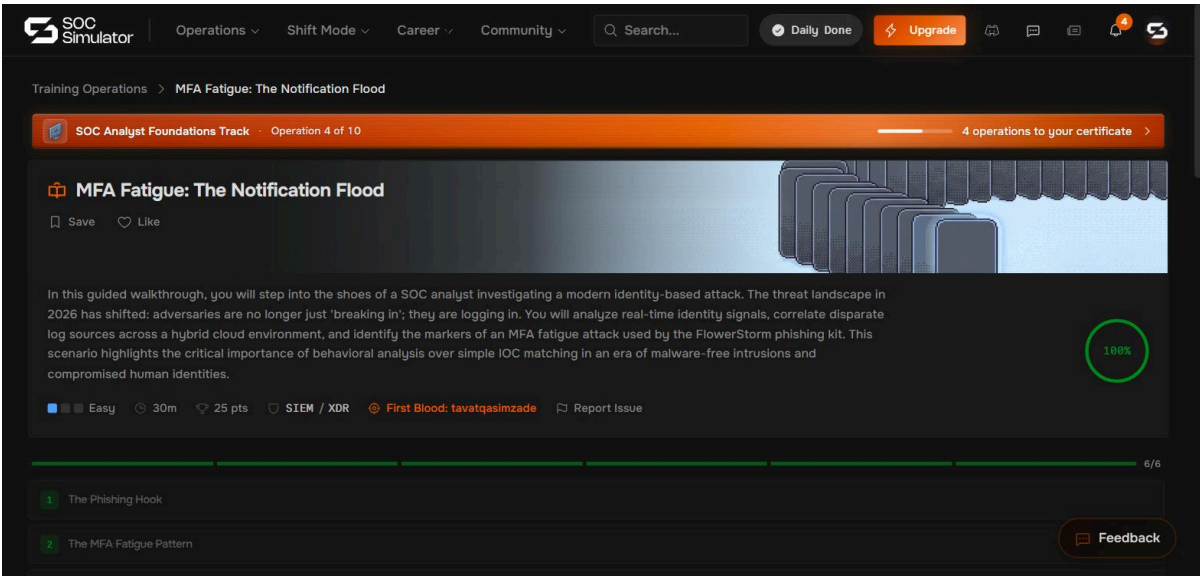


Figure 2. The XDR event timeline: the fake Dropbox email from dropbox-notify@drbx-share.com at 14:22:05 and the High-severity redirect drbx-share.com → flowerstorm-auth.net (AiTM proxy detected) at 14:25:18.

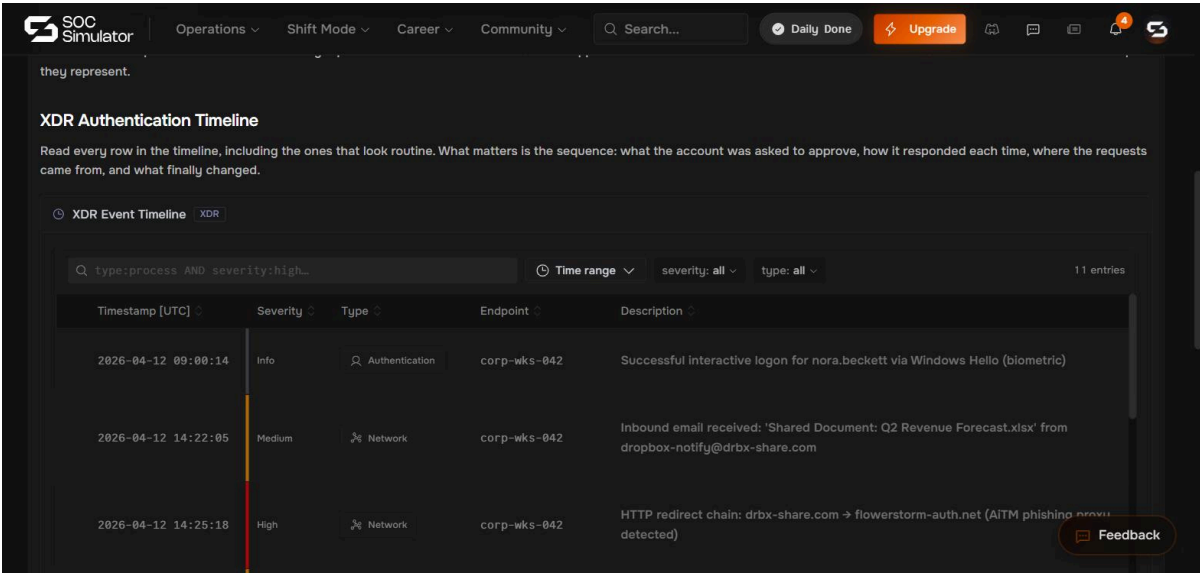


Figure 3. The authentication timeline: repeated MFA push prompts to nora.beckett DENIED, then APPROVED at 14:27:31 (Critical).

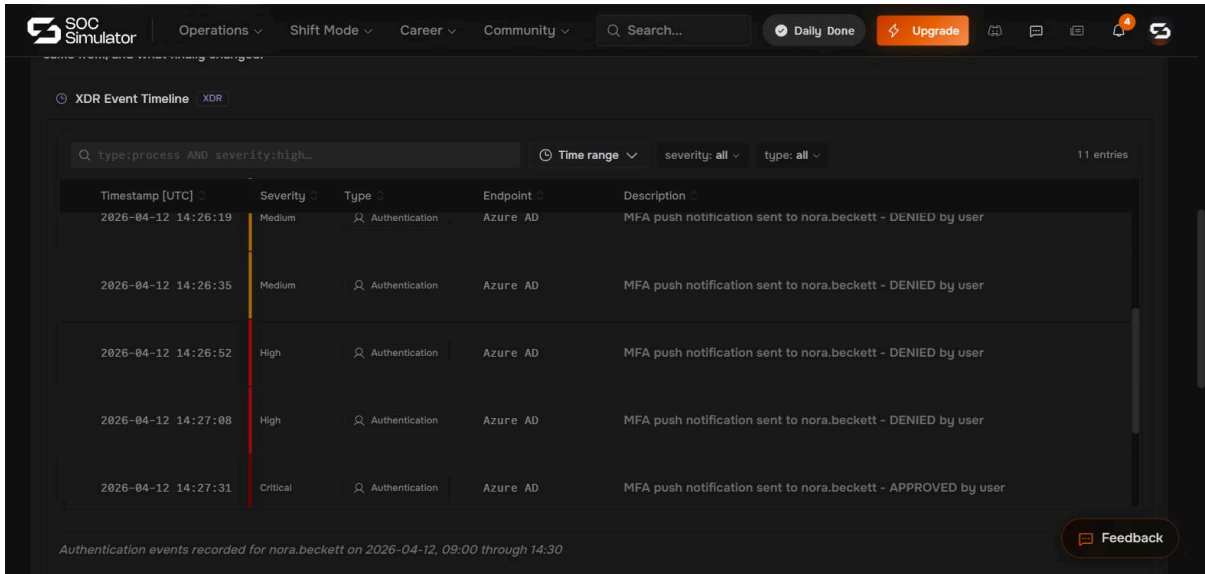


Figure 4. The authentication timeline, final rows: the approval at 14:27:31 followed by a successful token-based logon from 198.88.230.252 that dropped adversary-ai-agent.exe beaconing to 103.112.22.236.

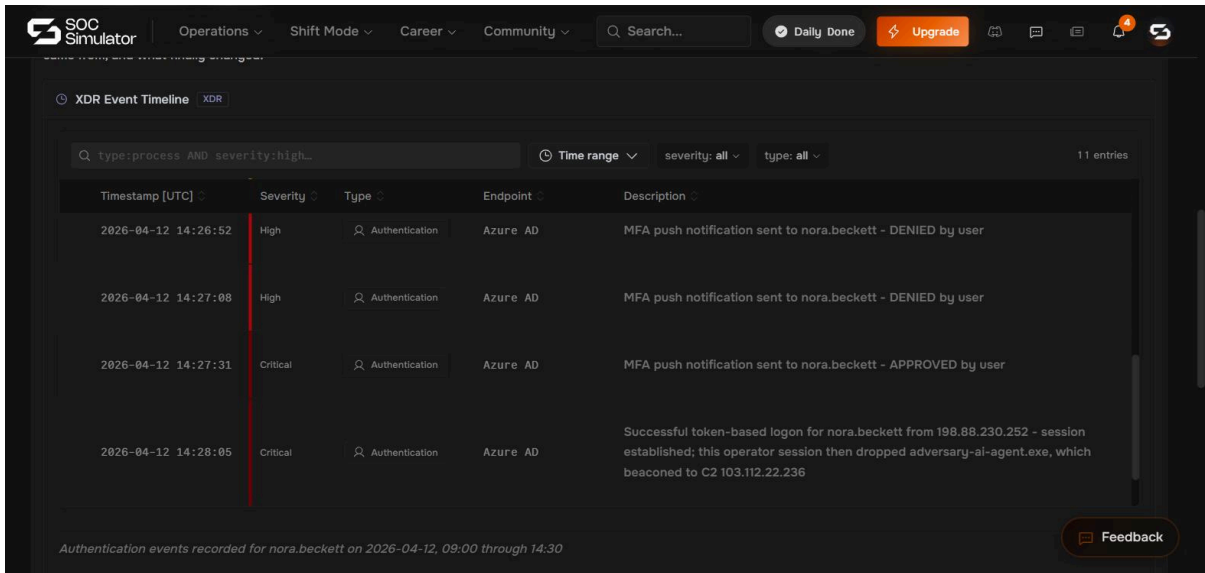


Figure 5. The Task 2 “Confirmed finding” panel: t1621 (Multi-Factor Authentication Request Generation).

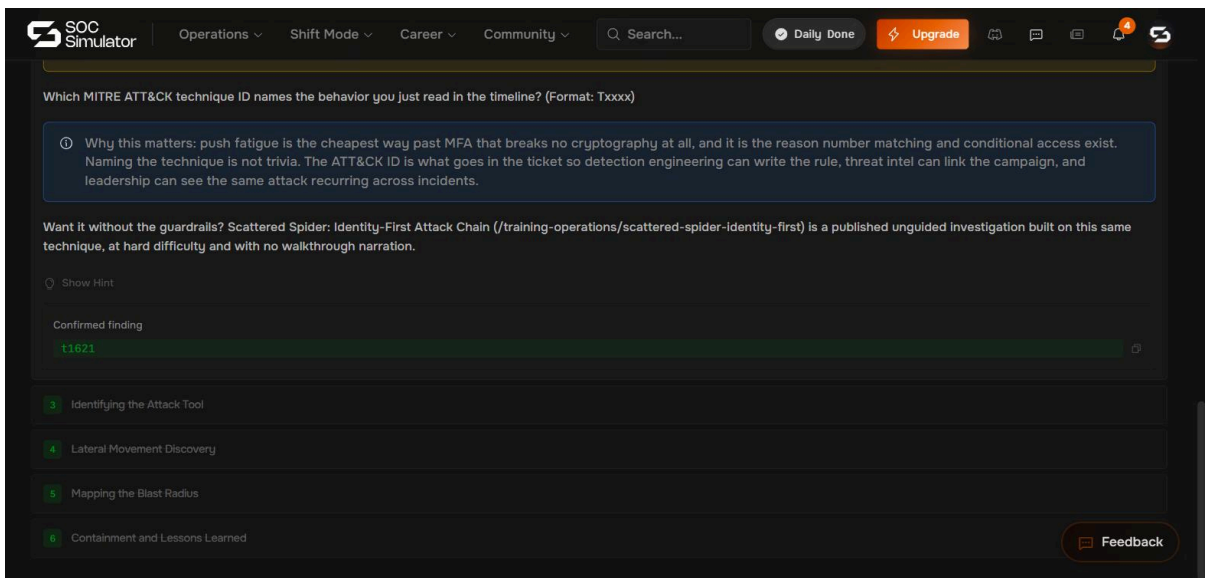


Figure 6. The XDR process tree for corp-wks-042: adversary-ai-agent.exe (Suspicious) talking to c2.flowerstorm-auth.net:8443, with child processes cmd.exe (recon), python.exe running wmiexec.py against 10.0.1.112, and certutil.exe downloading python-embed.zip.

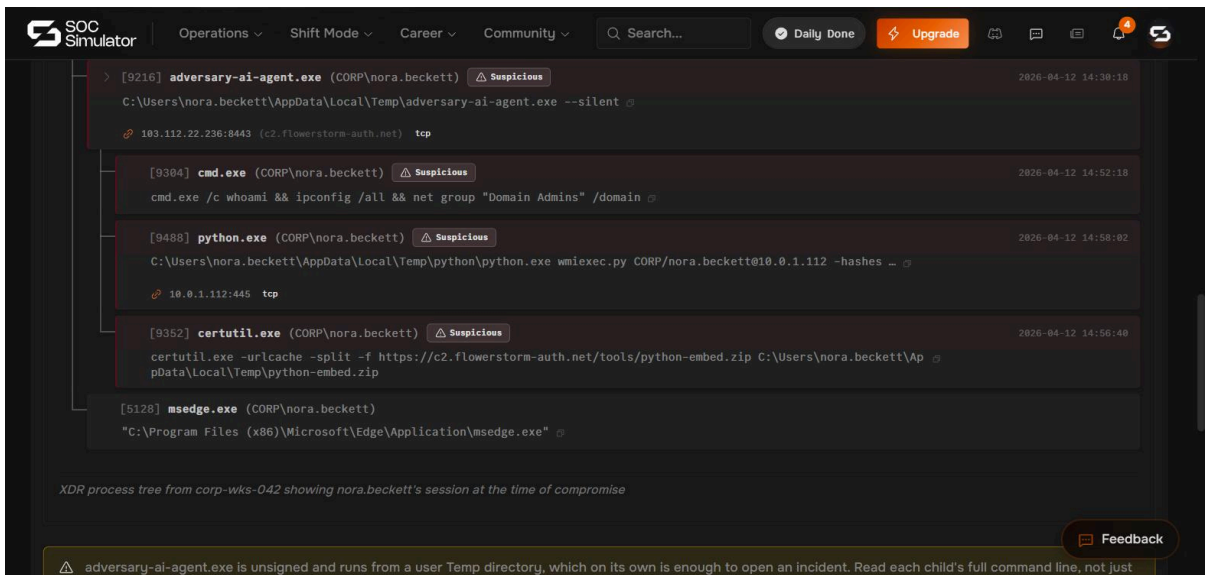


Figure 7. The Task 3 “Confirmed finding” panel: python.exe as the process that directly executed wmiexec.py.

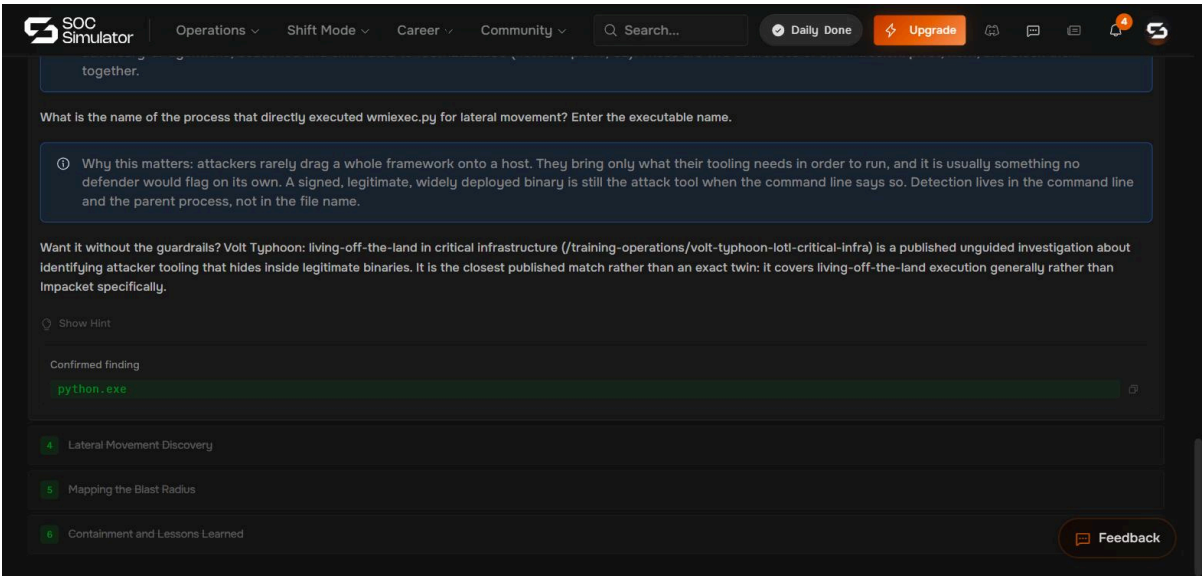


Figure 8. The SIEM log row at 14:58:14 expanded to its parsed fields: CommandLine “python.exe wmiexec.py CORP/nora.beckett@10.0.1.112 -hashes aad3b435...:ccef208a...”, destination 10.0.1.112, internal zone.

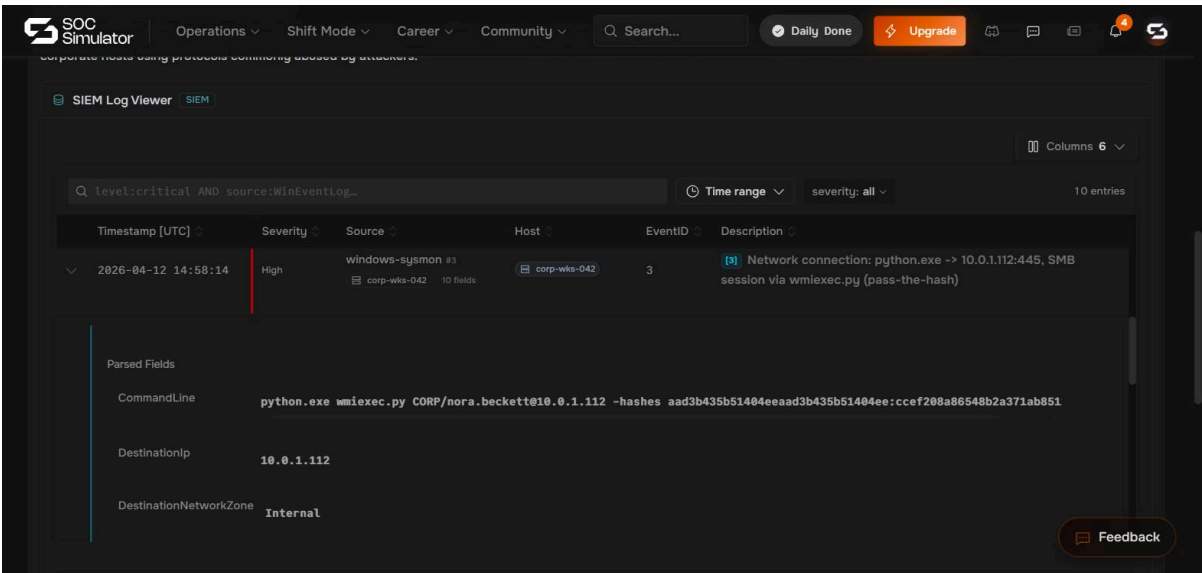


Figure 9. The SIEM network log's last rows: failed SMB attempt to 10.0.1.98 and, at 23:45:33, adversary-ai-agent.exe sending 4.2 GB over 6 hours to 103.112.22.236 (High).

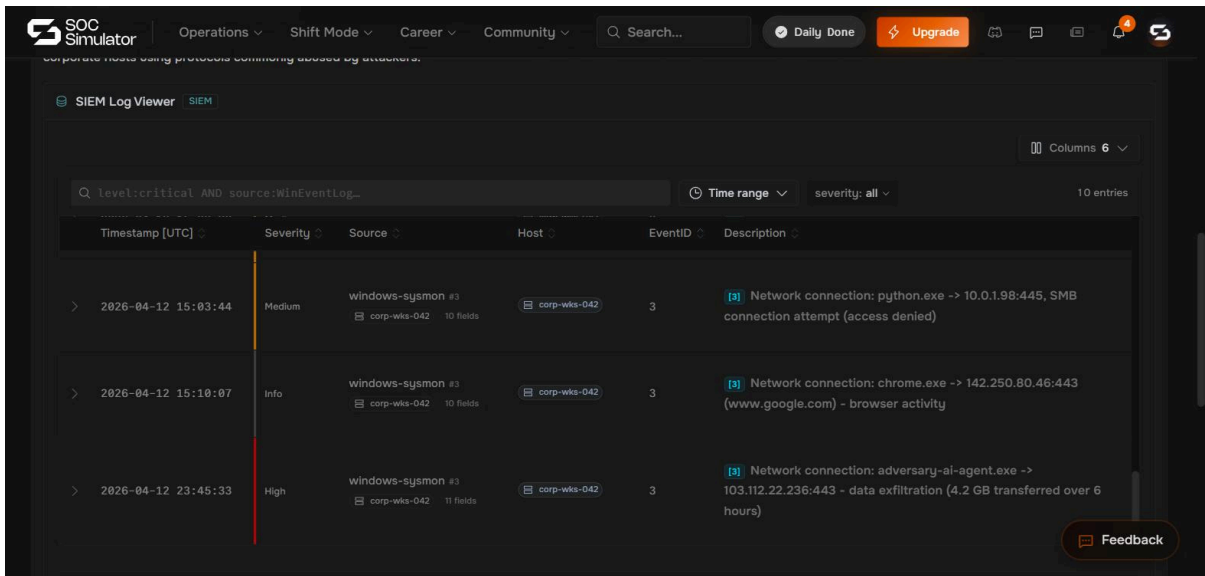


Figure 10. The XDR lateral-movement timeline: SMB access to WS-HR-01 and WS-SALES-01 FAILED (access denied), while WS-MKT-02 shows file staging to C:\Windows\Temp\staging\.

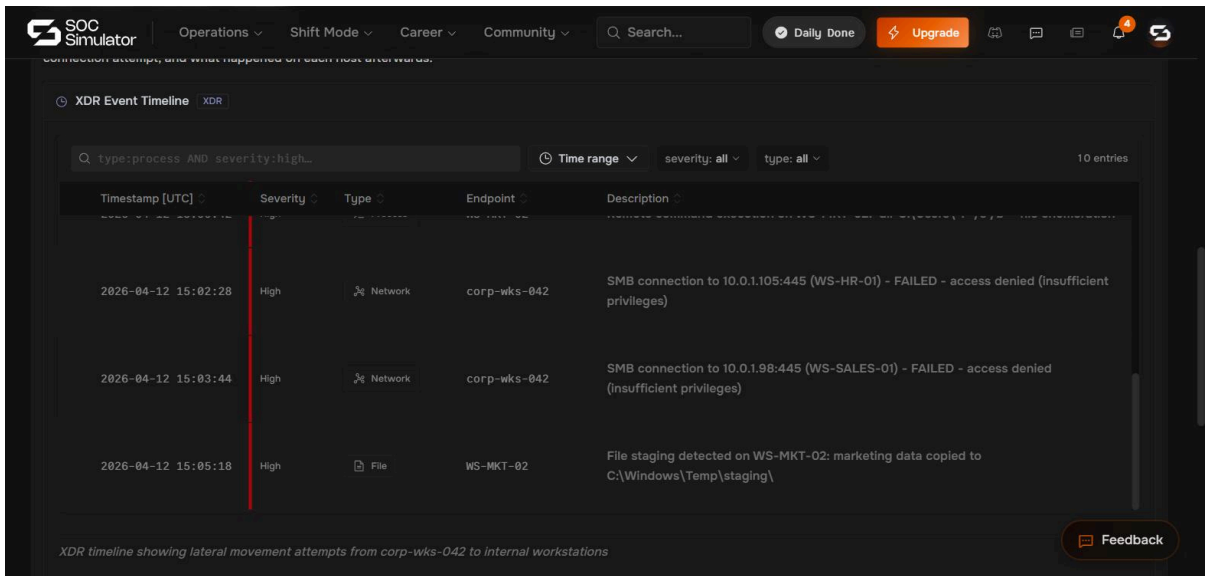


Figure 11. The platform's attack chain summary and immediate containment actions, confirming T1621, T1021.002, WS-MKT-02, the NTLM hash and the 4.2 GB exfiltration.

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Attack Chain Summary

1. Nora Beckett received a phishing email at 14:22 disguised as a Dropbox file-sharing notification from `dropbox-notify@drbx-share.com`
2. At 14:25 she clicked the link, which redirected through `drbx-share.com` to `flowerstorm-auth.net`, an AITM phishing proxy that captured her credentials and session cookie
3. At 14:26 the attacker triggered an MFA fatigue attack (T1621) from IP 198.88.230.252: five denied push notifications followed by one approval at 14:27
4. At 14:30 the attacker deployed `adversary-ai-agent.exe` to `nora.beckett's` Temp folder via the compromised session, an unsigned executable that established C2 communication with 103.112.22.236
5. The agent used `certutil.exe` (LOLBin) to download a portable Python runtime, then executed `wmlxec.py` (`python.exe`) for lateral movement via SMB (T1021.002) to WS-MKT-02 (10.0.1.112) using pass-the-hash
6. The attacker staged 4.2 GB of marketing data on WS-MKT-02 and exfiltrated it to 103.112.22.236 over 6 hours, completing the exfiltration at 23:45

Immediate Containment Actions

- Isolate `corp-wks-042` (10.0.2.219) and WS-MKT-02 (10.0.1.112) from the network via XDR endpoint isolation
- Block 198.88.230.252, 103.112.22.236, and `flowerstorm-auth.net` at the perimeter firewall and DNS filtering
- Revoke all active sessions and tokens for `nora.beckett`; force password reset and MFA re-enrollment
- Invalidate the NTLM hash `aad3b435b51404eeaad3b435b51404ee:cccf208a86548b2a371ab851255ff093` used in pass-the-hash; force credential rotation
- Search for `adversary-ai-agent.exe` file hash across all endpoints and hunt for `wmlxec.py` execution artifacts
- Audit WS-HR-01 and WS-SALES-01: the attacker attempted access but was denied, verify no other paths were used
- Notify the data protection team about the 4.2 GB exfiltration from the marketing department for potential regulatory reporting

Feedback

Figure 12. VirusTotal lookup for `flowerstorm-auth.net`: “Item not found” (never submitted).

URL, IP address, domain or file hash | Sign in | Sign up

Item not found

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Figure 13. VirusTotal lookup for drbx-share.com: “Item not found” (never submitted).

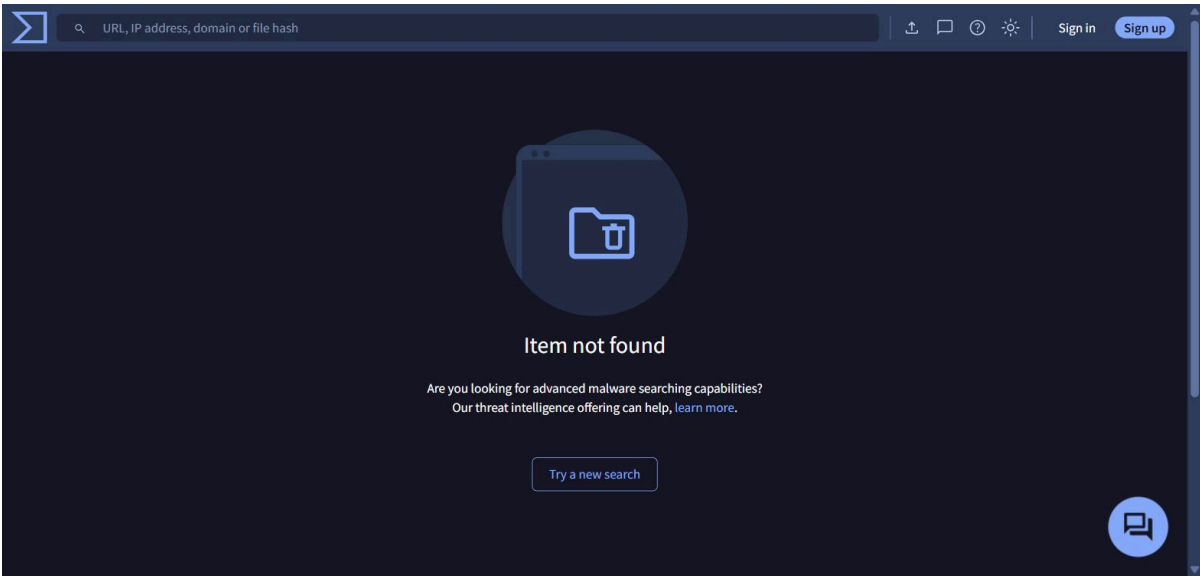


Figure 14. VirusTotal results for 198.88.230.252: 0 of 89 vendors flag it (AS 2914, NTT America, US).

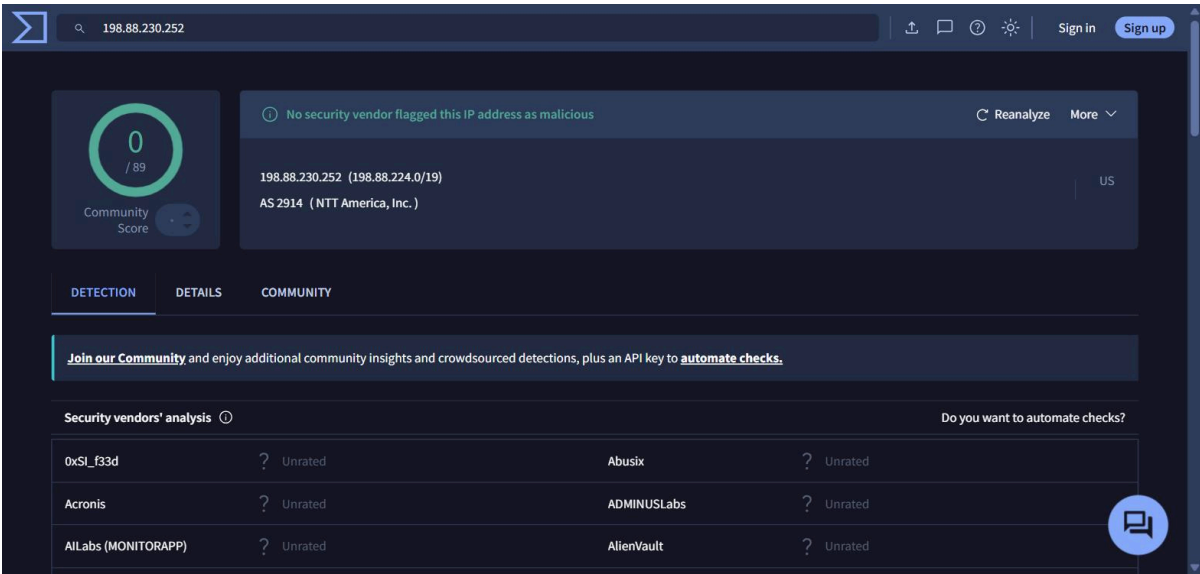


Figure 15. VirusTotal results for 103.112.22.236: 0 of 89 vendors flag it (AS 137678, Protoact Digital Network, IN).

